

Sample Worksheets

The printouts of the PreTeXt sample article

1 Worksheets

This is a section full of worksheets. Each is a division of its own, via the `<worksheet>` element. This is an optional `<introduction>` to the current `<section>`. In practice you might want to rip out all the worksheets of an entire book and bundle them up as an “activity book.”

If you make PDF output you will notice an increased amount of control over layout. Also, if the publication file elects L^AT_EX draft mode, then there will be visual indicators of prescribed whitespace.

1.1 A Geometric Prelude (without authored pages)

Objectives

- Practice visualizing vector addition
- Use vectors without explicit coordinates

This two-page worksheet was generously donated to the sample article by Dave Rosoff at a CuratedCourses workshop in August 2018. It has the default (skinny) margins.

It was known to Euclid, and probably earlier, that the midpoints of the sides of any quadrilateral all lie in the same plane (even if the vertices of the quadrilateral do not). In fact, these midpoints are the vertices of a parallelogram, as pictured in [Figure 1.1](#).

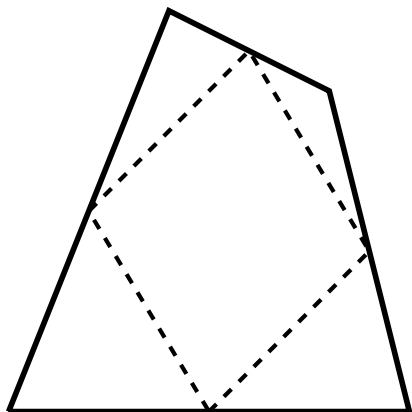


Figure 1.1 The midpoints of the sides of a quadrilateral are the vertices of a parallelogram.

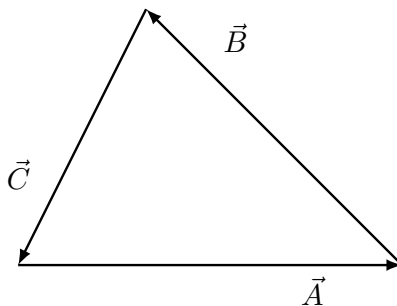


Figure 1.2 The sides of a triangle presented as vectors.

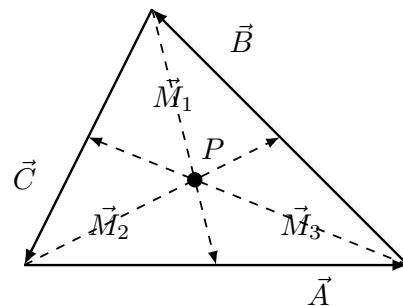


Figure 1.3 The medians of the triangle are $\vec{M}_1$, $\vec{M}_2$, and $\vec{M}_3$.

In this exercise, we'll use vectors to show that the medians of any triangle ([Figure 1.2](#)) intersect at a point. Recall that medians are the lines connecting the vertices of the triangle to the midpoints of their opposite edges, as in the figure. We'll do this in a few steps.

1. What is the value of $\vec{A} + \vec{B} + \vec{C}$?

Figure 1.3 from the previous page is reproduced for your convenience.

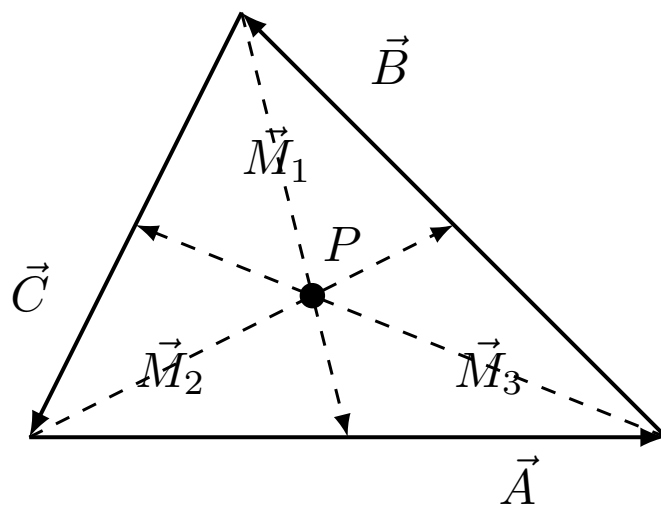


Figure 1.4 The medians of the triangle are $\vec{M}_1$, $\vec{M}_2$, and $\vec{M}_3$.

2. Show that $\vec{M}_1 + \vec{M}_2 + \vec{M}_3 = 0$.

Hint. Use [Worksheet Exercise 1.1.1](#).

3. To show that the point P exists (as the common intersection of the $\vec{M}_i$), show that

$$\vec{A} + \frac{2}{3}\vec{M}_3 = \frac{2}{3}\vec{M}_2 = \boxed{}.$$

4. If you have time, try to devise a vector proof of Euclid's result presented at the beginning of the workshop. Recall that a **parallelogram** is a four-sided polygon whose opposite sides are parallel.

Wrap-up. It's possible to do interesting things with vector arithmetic in a coordinate-free way: we didn't specify an origin, or any entries of any vectors in the examples.

1.2 A Geometric Prelude (with authored pages)

Objectives

- Practice visualizing vector addition
- Use vectors without explicit coordinates

This two-page worksheet was generously donated to the sample article by Dave Rosoff at a CuratedCourses workshop in August 2018. It has the default (skinny) margins.

It was known to Euclid, and probably earlier, that the midpoints of the sides of any quadrilateral all lie in the same plane (even if the vertices of the quadrilateral do not). In fact, these midpoints are the vertices of a parallelogram, as pictured in [Figure 1.1](#).

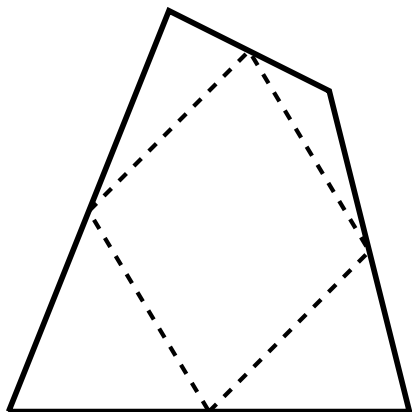


Figure 1.5 The midpoints of the sides of a quadrilateral are the vertices of a parallelogram.

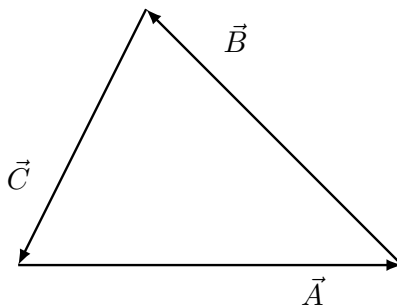


Figure 1.6 The sides of a triangle presented as vectors.

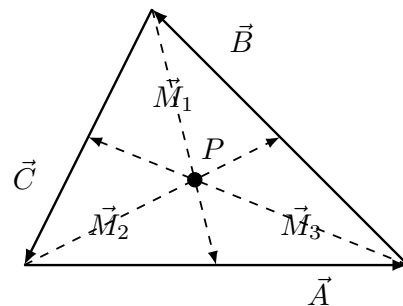


Figure 1.7 The medians of the triangle are $\vec{M}_1$, $\vec{M}_2$, and $\vec{M}_3$.

In this exercise, we'll use vectors to show that the medians of any triangle ([Figure 1.2](#)) intersect at a point. Recall that medians are the lines connecting the vertices of the triangle to the midpoints of their opposite edges, as in the figure. We'll do this in a few steps.

1. What is the value of $\vec{A} + \vec{B} + \vec{C}$?

Figure 1.3 from the previous page is reproduced for your convenience.

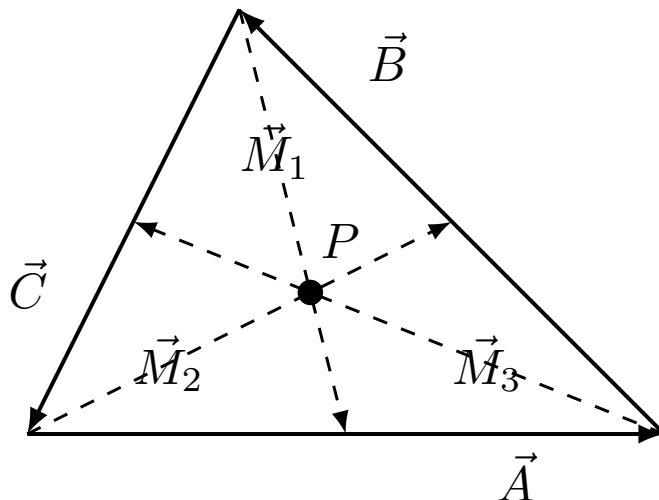


Figure 1.8 The medians of the triangle are $\vec{M}_1$, $\vec{M}_2$, and $\vec{M}_3$.

2. Show that $\vec{M}_1 + \vec{M}_2 + \vec{M}_3 = 0$.

Hint. Use [Worksheet Exercise 1.1.1](#).

3. To show that the point P exists (as the common intersection of the $\vec{M}_i$), show that

$$\vec{A} + \frac{2}{3}\vec{M}_3 = \frac{2}{3}\vec{M}_2 = \text{[gray box]}.$$

4. If you have time, try to devise a vector proof of Euclid's result presented at the beginning of the workshop. Recall that a **parallelogram** is a four-sided polygon whose opposite sides are parallel.

Wrap-up. It's possible to do interesting things with vector arithmetic in a coordinate-free way: we didn't specify an origin, or any entries of any vectors in the examples.

1.3 Networks Worksheet (no authored pages)

Basic laws for electrical circuits. This two-page worksheet was generously donated to the sample article by Virgil Pierce at a CuratedCourses workshop in August 2018. It has default (skinny) left and right margins, but we have specified longer top and bottom margins, with the top being the larger of the two.

Theorem 1.9 Ohms Law. *The current through a resistor is proportional to the ratio of the Voltage to the Resistance*

$$I = \frac{V}{R}$$

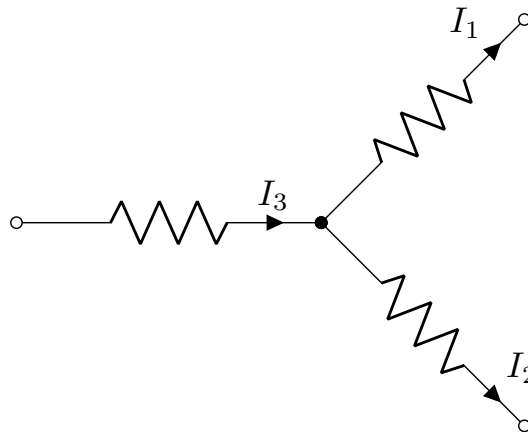
Or for our purposes

$$IR = V$$

Theorem 1.10 Kirchoffs Current Law. *The sum of the currents in a network meeting at a point is zero.*

$$\sum_{k=1}^n I_k = 0$$

Example 1.11 Kirchoff's Current Law. For the circuit below $I_1 + I_2 = I_3$.

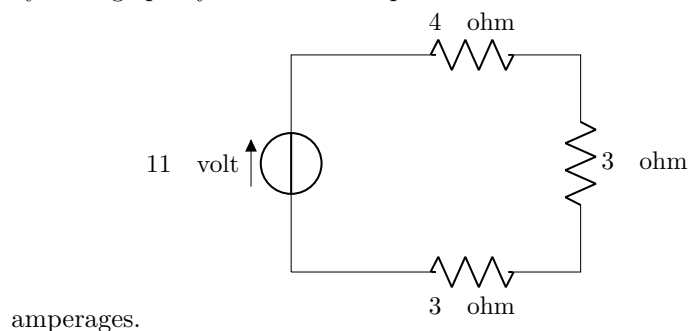


Theorem 1.12 Kirchoffs Voltage Law. *The sum of the voltages around any closed circuit (or subcircuit) is zero.* ▲

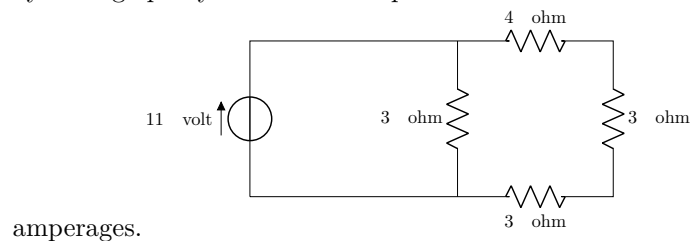
$$\sum_{k=1}^n V_k = 0$$

Kirchoffs Current Law and Kirchoffs Voltage Law combined with Ohms Law gives for any circuit of resistors and sources a linear system that may (or may not) determine the currents.

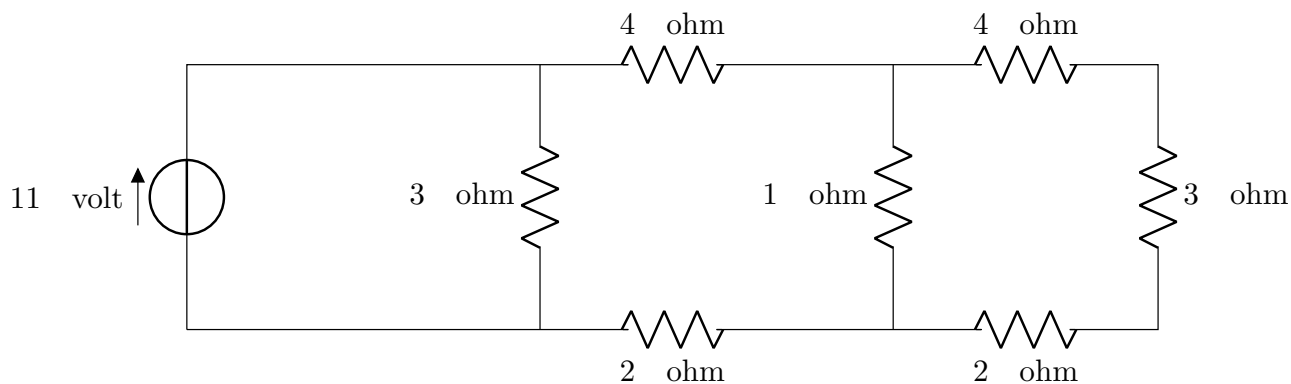
1. For the simple network pictured, calculate the amperage in each part of the network by setting up a system of linear equations for the



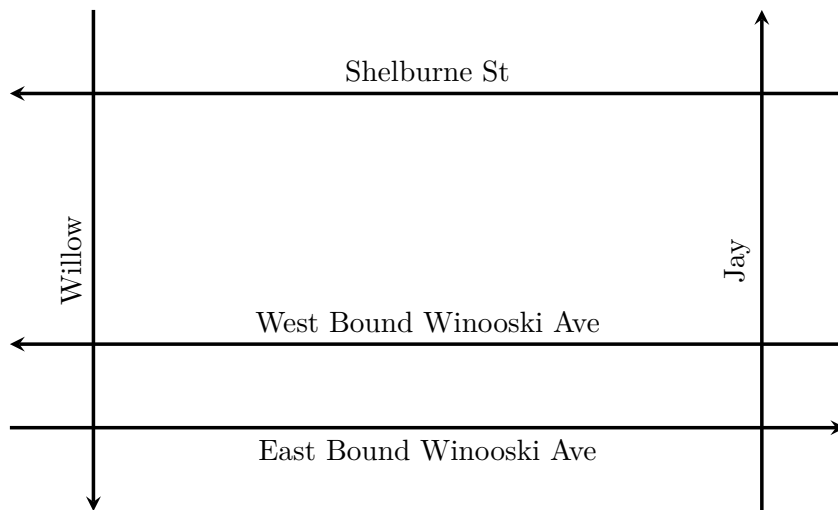
2. Compare it with a parallel circuit network. Calculate the amperage in each part of the network by setting up a system of linear equations for the



3. Now for a more complicated network. Calculate the amperage in each part of the network by setting up a system of linear equations for the amperages.



4. Now generalize these ideas to a context outside of electrical circuits. Consider the network of streets given in the diagram (with one-way directions as indicated).



A traffic engineer counts the hourly flow of cars into and out of this network at the entrances. They get (EB = East Bound; WB = West Bound):

Table 1.13 Estimated hourly traffic flow for the road network

	EB Winooski	WB Winooski	Shelburne St	Willow	Jay
into	50	400	0	10	50
out of	55	390	20	15	30

Use a variable for each segment inside of the network and set up a system of linear equations restricting the flow. Solve the system. Note that you should not get a unique solution as traffic should be able to flow through the network in various ways.

1.4 Networks Worksheet (with authored pages)

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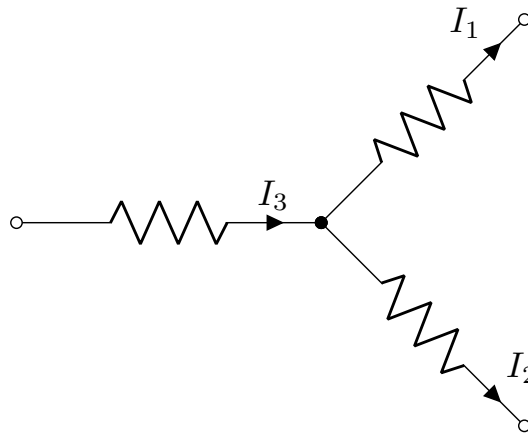
Or for our purposes

$$IR = V$$

Theorem 1.15 Kirchoffs Current Law. *The sum of the currents in a network meeting at a point is zero.*

$$\sum_{k=1}^n I_k = 0$$

Example 1.16 Kirchoff's Current Law. For the circuit below $I_1 + I_2 = I_3$.

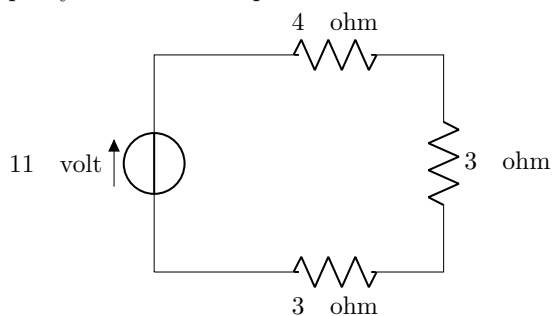


Theorem 1.17 Kirchoffs Voltage Law. *The sum of the voltages around any closed circuit (or subcircuit) is zero.* ▲

$$\sum_{k=1}^n V_k = 0$$

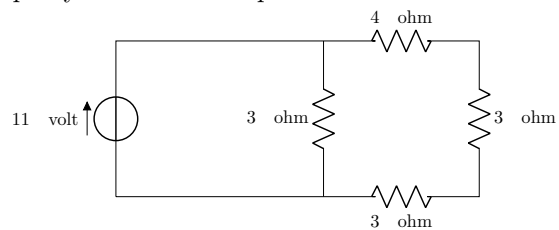
Kirchoffs Current Law and Kirchoffs Voltage Law combined with Ohms Law gives for any circuit of resistors and sources a linear system that may (or may not) determine the currents.

1. For the simple network pictured, calculate the amperage in each part of the network by setting up a system of linear equations for the



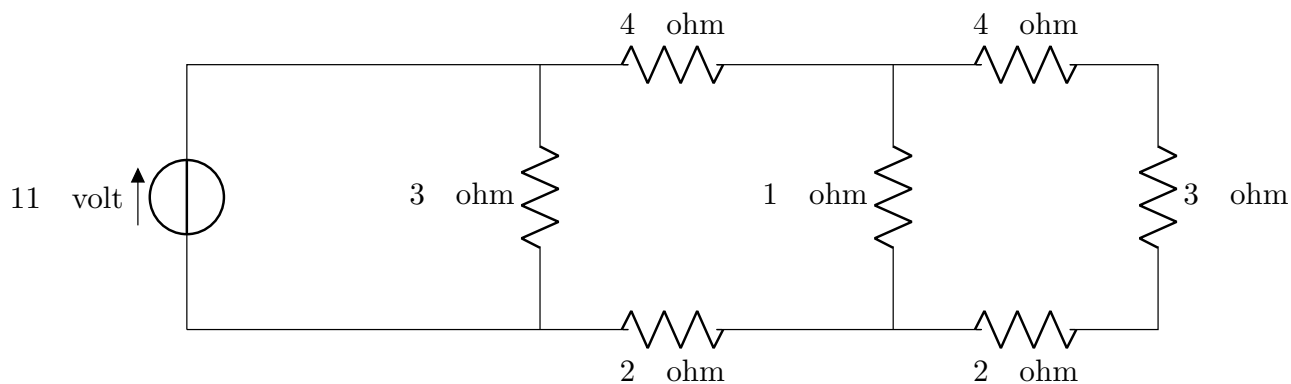
amperages.

2. Compare it with a parallel circuit network. Calculate the amperage in each part of the network by setting up a system of linear equations for the

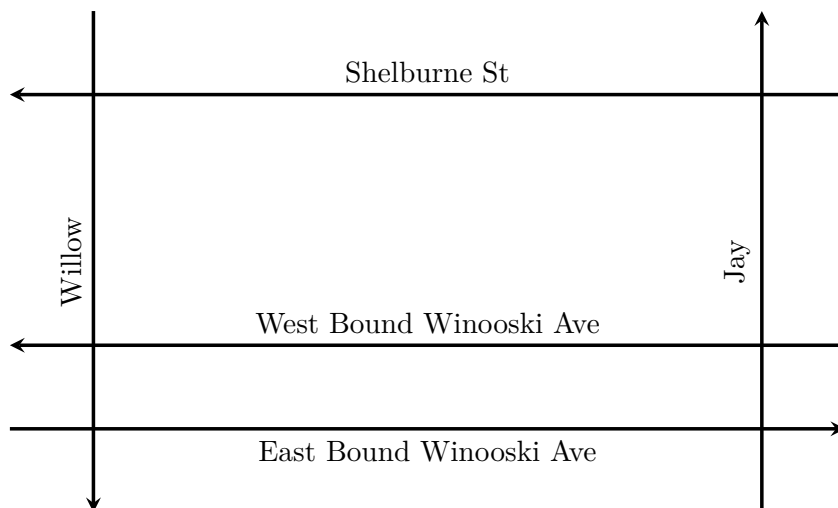


amperages.

3. Now for a more complicated network. Calculate the amperage in each part of the network by setting up a system of linear equations for the amperages.



4. Now generalize these ideas to a context outside of electrical circuits. Consider the network of streets given in the diagram (with one-way directions as indicated).



A traffic engineer counts the hourly flow of cars into and out of this network at the entrances. They get (EB = East Bound; WB = West Bound):

Table 1.18 Estimated hourly traffic flow for the road network

	EB Winooski	WB Winooski	Shelburne St	Willow	Jay
into	50	400	0	10	50
out of	55	390	20	15	30

Use a variable for each segment inside of the network and set up a system of linear equations restricting the flow. Solve the system. Note that you should not get a unique solution as traffic should be able to flow through the network in various ways.

1.5 Worksheet

This is a mock one-page worksheet for testing purposes. We have specified an overall margin just slightly less than the default.

1. Praesent rutrum scelerisque felis sit amet adipiscing. Phasellus in mollis velit. Nunc malesuada felis sit amet massa cursus, eget elementum neque viverra.

2. Integer sagittis dictum turpis vel aliquet. Fusce ut suscipit dolor, nec tristique nisl. Aenean luctus, leo et ornare fermentum, nibh dui vulputate leo, nec tincidunt augue ipsum sed odio. Nunc non erat sollicitudin, iaculis eros consequat, dapibus eros.

Hint. This is a hint. You can hide it in the print preview before printing.

Answer. Here is an answer. Also hidable.

Solution. This is a rather long solution. When you hide or show it in the print preview, the workspace should be adjusted (the page will reload).

Integer sagittis dictum turpis vel aliquet. Fusce ut suscipit dolor, nec tristique nisl. Aenean luctus, leo et ornare fermentum, nibh dui vulputate leo, nec tincidunt augue ipsum sed odio. Nunc non erat sollicitudin, iaculis eros consequat, dapibus eros.

A two-line paragraph interspersed to check on spacing, breaks and all that.

3. **A full-width exercise.** Praesent rutrum scelerisque felis sit amet adipiscing. Phasellus in mollis velit. Nunc malesuada felis sit amet massa cursus, eget elementum neque viverra.

Integer sagittis dictum turpis vel aliquet. Fusce ut suscipit dolor, nec tristique nisl. Aenean luctus, leo et ornare fermentum, nibh dui vulputate leo, nec tincidunt augue ipsum sed odio. Nunc non erat sollicitudin, iaculis eros consequat, dapibus eros.

Solution. This is a sample solution for a full-width exercise. Integer sagittis dictum turpis vel aliquet. Fusce ut suscipit dolor, nec tristique nisl. Aenean luctus, leo et ornare fermentum, nibh dui vulputate leo, nec tincidunt augue ipsum sed odio. Nunc non erat sollicitudin, iaculis eros consequat, dapibus eros.

Another two-line paragraph interspersed to check on spacing, breaks and all that.

- | | | |
|--|---|--|
| <ol style="list-style-type: none">4. Praesent rutrum scelerisque felis sit amet adipiscing. Phasellus in mollis velit. Nunc malesuada felis sit amet massa cursus, eget elementum neque viverra. | <ol style="list-style-type: none">5. Integer sagittis dictum turpis vel aliquet. Fusce ut suscipit dolor, nec tristique nisl. Aenean luctus, leo et ornare fermentum, nibh dui vulputate leo, nec tincidunt augue ipsum sed odio. Nunc non erat sollicitudin, iaculis eros consequat, dapibus eros. | <ol style="list-style-type: none">6. Praesent rutrum scelerisque felis sit amet adipiscing. Phasellus in mollis velit. Nunc malesuada felis sit amet massa cursus, eget elementum neque viverra. |
|--|---|--|

Activity 1.19 A Mock Activity. The problem, as we see it.

Hint. A worksheet could have hints, no? But no spacing. Note row below has widths set to balance the heights.

7. Praesent rutrum scelerisque felis sit amet adipiscing. Phasellus in mollis velit. Nunc malesuada felis sit amet massa cursus, eget elementum neque viverra.
8. Integer sagittis dictum turpis vel aliquet. Fusce ut suscipit dolor, nec tristique nisl. Aenean luctus, leo et ornare fermentum, nibh dui vulpate leo, nec tincidunt augue ipsum sed odio. Nunc non erat sollicitudin, iaculis eros consequat, dapibus eros.
9. Praesent rutrum scelerisque felis sit amet adipiscing. Phasellus in mollis velit. Nunc malesuada felis sit amet massa cursus, eget elementum neque viverra.

1.6 Dot products and projection

1. Let $\vec{v}_1 = (-4, 1)$, $\vec{v}_2 = (2, 2)$, $\vec{v}_3 = (1, 2, 3)$, $\vec{v}_4 = (-2, 1, 0)$. Find the values of the following expressions:

(a) $\vec{v}_1 \cdot \vec{v}_2 =$

(b) $\vec{v}_3 \cdot \vec{v}_4 =$

(c) $\|\vec{v}_1\| =$

(d) $\|\vec{v}_4\| =$

- (e) Are any of these vectors perpendicular to each other? _____

2. The vectors $\vec{a} = (3, 9)$ and $\vec{b} = (4, 2)$ are pictured below. Derive the formula for projection on a line and use it to find the projection of $\vec{a}$ on the line spanned by $\vec{b}$. Also compute the length of the residual vector.¹

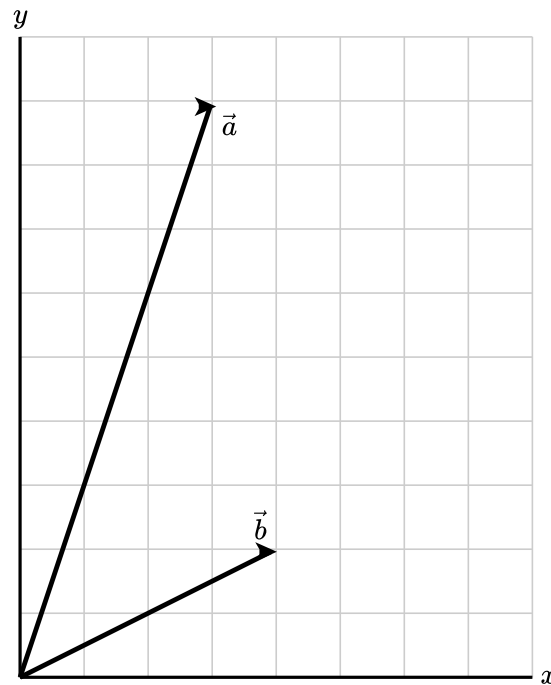


Figure 1.20 Two vectors $\vec{a}$ and $\vec{b}$.

¹Here is a footnote.

3. Consider the vector equation

$$m \begin{bmatrix} 2 \\ 5 \end{bmatrix} = \begin{bmatrix} 3 \\ 7 \end{bmatrix}.$$

(a) Check that there is no solution m that makes the equation true.

(b) Use projection to find the best approximation $\hat{m}$.

(c) Compute $\hat{m} \begin{bmatrix} 2 \\ 5 \end{bmatrix}$.

(d) Compute the residual vector.

(e) Compute the length of the residual vector and explain what it means.

4. Consider the system of equations

$$3t = 5$$

$$2t = 9.$$

(a) Write the system in vector form.²

(b) Find the best estimate, $\hat{t}$, of t using projection.

(c) Compute the length of the residual vector.

²Another footnote

1.7 Worksheet

Activity 1.21 Just a simple activity here.

Activity 1.22 Here is a second activity.

1.8 Worksheet

Activity 1.23 This is going to be an activity with tasks.

(a) Here is the first task.

Solution. Here is a solution for the first task. It should get indented the same amount as the task.

(b) Here is the second task.

(c) Here is the third task.

This is a conclusion that comes after the last task.

1.9 Worksheet

This is a mock worksheet for testing the use of `@workspace` on an `<exercise>` in a worksheet.

Do things to the following.

1. Apple
2. Banana
3. Cherry
4. Durian
5. Elderberry (with workspace override)
6. Fig
7. Guava

8. Habanero

9. I can't think of an "I" fruit.

10. Jackfruit

Now, do things to the following, again.

11. Apple

12. Banana

13. Cherry

14. Durian

15. Elderberry (with workspace override)

16. Fig

17. Guava

18. Habanero

19. I can't think of an "I" fruit.

20. Jackfruit

1.10 Worksheet with an exercise with tasks and subtasks

Page breaking between tasks in a single exercise can be tricky. This is a test for that.

1. Here are multiple tasks with just enough workspace to cause a page break after the third one.

- (a) This task has two short subtasks that should be indented under it.

- (i) The first subtask.

- (ii) The second subtask.

- Conclusion for the first task.

- (b) The second task.

- (c) The third task.

- (d) The fourth task.

- Conclusion for the entire exercise.

2. An exercise without any tasks but some workspace.

1.11 Worksheet with <paragraphs>

This is a test of the <paragraphs> element within a worksheet. It should be able to contain exercises that have workspace, and we should be able to insert page breaks even inside of paragraphs.

Warm up.

1 1.24 section-worksheets-13-3-2 This is a warm-up exercise. It should have a small workspace and be on the first page of the worksheet.

More exercises.

2 1.25 section-worksheets-13-4-2 This is a second exercise. It should have a larger workspace and be on the second page of the worksheet.

3 1.26 section-worksheets-13-4-3 This is a third exercise with some tasks. The workspace should be such that there will be a page break between tasks.

(a) Do the first thing.

(b) Do the second thing.

(c) Do the third thing.

A third division. This is a paragraph that starts the third <paragraphs> division.

4. This is an exercise in the third paragraphs division inside a side-by-side.
5. This is a second exercise in the third paragraphs division. It should have a larger workspace and be on the fourth page of the worksheet.

6 1.27 section-worksheets-13-5-4 This is a third exercise in the third paragraphs division. It should have a larger workspace and be on the fifth page of the worksheet.

2 Handouts

Like worksheets, a **handout** is a division that is intended to be printed for use in a classroom. In HTML output, you get the same print preview and page layout as you do with worksheets; in PDF, these will start on a new page with possibly different margins than the rest of the document.

Unlike worksheets, handouts do not have a special class of exercises or activities (exercises in a handout are treated like an *inline* exercise). The other main distinction is that a handout lets workspace be specified on pretty much any block or paragraph element, not just exercises, tasks, and project-like elements. This allows them to be used in the creation of guided notes containing some premade content with lots of space for students to fill in details during class.

2.1 Derivative Rules

Rules for specific types of functions.		Constant functions	$(k)' = 0$
Power functions	$(x^n)' = nx^{n-1}$		
Exponential functions	$(a^x)' = \ln(a)a^x$ (for $a > 0$) $(e^x)' = e^x$		
Logarithmic functions	$(\ln(x))' = \frac{1}{x}$		
Trigonometric functions	$(\sin(x))' = \cos(x).$ $(\cos(x))' = -\sin(x).$ $(\tan(x))' = \sec^2(x) = \frac{1}{\cos^2(x)}.$ $(\arcsin(x))' = \frac{1}{\sqrt{1-x^2}}.$ $(\arctan(x))' = \frac{1}{1+x^2}.$		
Rules for combinations of functions.		Constant multiples	$(k \cdot f(x))' = k \cdot f'(x)$
Sum and difference	$(f(x) \pm g(x))' = f'(x) \pm g'(x)$		
Products (the product rule)	$(f(x) \cdot g(x))' = f'(x)g(x) + f(x)g'(x)$		
Quotients (the quotient rule)	$\left(\frac{f(x)}{g(x)}\right)' = \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2}$		
Compositions (the chain rule)	$(f(g(x)))' = f'(g(x)) \cdot g'(x)$		

2.2 Guided Notes: Derivatives of sums

Today we will explore how to take the derivative of the sum of two functions. For example, if

$$f(x) = x^2 + 3x,$$

what is $f'(x)$? What two functions is this the sum of? What are the derivatives of each of those functions?

To be sure of the derivative of the sum, we should use the definition of the derivative.

Definition 2.1 Definition of the Derivative. The derivative of a function $f(x)$ at any point x is defined as...

Now let's apply this definition to the function $f(x) = x^2 + 3x$. We have:



What should the general rule be?

Theorem 2.2 For any two differentiable functions $f(x)$ and $g(x)$, the derivative of their sum $h(x) = f(x) + g(x)$ is given by:

Proof. Let $f(x)$ and $g(x)$ be differentiable functions and let $h(x) = f(x) + g(x)$. Then by the limit definition of the derivative,

Example 2.3 Find the derivative of $f(x) = x^5 + e^x$. ■

Example 2.4 Find the derivative of $f(x) = \sqrt{x} + x^3 + 7x$. ▲

Example 2.5 Find the derivative of $f(x) = 5x^4$. ▲

▲

2.3 A handout with a long list

It is common for a handout, especially a set of lecture notes, to consist of a long list of bullets, possibly more than would fit on a page. This handout should test that such a handout can have its items split between pages even if they belong to the same paragraph.

- This is the first item in a long list. It should be on the first page of the handout.
- This is the second item in a long list. It should be on the first page of the handout.
- This is the third item in a long list. It should be on the second page of the handout.

Maybe we have a second paragraph in the same list item. This is really just to help fill up space. Maybe I should have used some of that standard space filler, but I'm just typing now because I can't remember what is call.

- I suppose I could just start copy/pasting now.
- Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.
- Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.
- Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.
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- See, I figured it out!

And that's the end of the list. Did this go onto a second page?

¹A late footnote

3 Projects as Printouts

A project-like element outside of any worksheet or handout may still carry **workspace**, on itself or on its tasks. When it does, HTML output gives it the same print preview icon a worksheet gets, and previewing it lays out just that project across pages with the workspace expanded to fill them. In PDF such a project is typeset as an ordinary block, with no extra room.

The third project below has no **workspace** anywhere, so it gets no icon.

Project 3.1 A Project With Workspace On Its Tasks. Show all of your reasoning in the space provided.

(a) Find the derivative of $f(x) = x^2$.

Hint. Use the power rule.

Solution. We have $f'(x) = 2x$.

(b) Now find the derivative of $g(x) = x^3$, and explain the pattern you see.

Compare your two answers with a neighbor.

Project 3.2 A Project With Workspace On Itself. Explain, in a paragraph, why the workspace requested here is not visible on screen, but does appear in the print preview.

Project 3.3 A Project With No Workspace. This project should not have a print preview icon.

4 Worksheet Solution Testing

Solutions to projects and worksheets can both be selected independently for the auto-generated solutions divisions. What happens when a worksheet contains a project? Here we explore the various combinations.

4.1 Subsection with only exercise

Checkpoint 4.1 This is an exercise's statement.

Answer. This is an exercise's answer.

Solution. This is the exercise's solution.

4.2 Subsection with only project

Project 4.2 This is a project's statement.

Answer. This is a project's answer.

Solution. This is the project's solution.

4.3 Subsection with both exercise and project

Checkpoint 4.3 This is an exercise's statement.

Answer. This is an exercise's answer.

Solution. This is the exercise's solution.

Project 4.4 This is a project's statement.

Answer. This is a project's answer.

Solution. This is the project's solution.

4.4 Worksheet with only exercise

1. This is an exercise's statement.

Answer. This is an exercise's answer.

Solution. This is the exercise's solution.

4.5 Worksheet with only project

Project 4.5 This is a project's statement.

Answer. This is a project's answer.

Solution. This is the project's solution.

4.6 Worksheet with both exercise and project

1. This is an exercise's statement.

Answer. This is an exercise's answer.

Solution. This is the exercise's solution.

Project 4.6 This is a project's statement.

Answer. This is a project's answer.

Solution. This is the project's solution.

4.7 Inline answers only

4.1 · Subsection with only exercise

Checkpoint 4.1 This is an exercise's answer.

4.3 · Subsection with both exercise and project

Checkpoint 4.3 This is an exercise's answer.

4.8 Project answers only

4.2 · Subsection with only project

Project 4.2 This is a project's answer.

4.3 · Subsection with both exercise and project

Project 4.4 This is a project's answer.

4.5 · Worksheet with only project

Project 4.5 This is a project's answer.

4.6 · Worksheet with both exercise and project

Project 4.6 This is a project's answer.

4.9 Worksheet answers only

4.4 · Worksheet with only exercise

4.4.1. This is an exercise's answer.

4.6 · Worksheet with both exercise and project

4.6.1. This is an exercise's answer.

4.10 Inline answers, project statement

4.1 · Subsection with only exercise

Checkpoint 4.1 Answer. This is an exercise's answer.

4.2 · Subsection with only project

Project 4.2 This is a project's statement.

4.3 · Subsection with both exercise and project

Checkpoint 4.3 Answer. This is an exercise's answer.

Project 4.4 This is a project's statement.

4.5 · Worksheet with only project

Project 4.5 This is a project's statement.

4.6 · Worksheet with both exercise and project

Project 4.6 This is a project's statement.

4.11 Inline answers, worksheet statement

4.1 · Subsection with only exercise

Checkpoint 4.1 Answer. This is an exercise's answer.

4.3 · Subsection with both exercise and project

Checkpoint 4.3 Answer. This is an exercise's answer.

4.4 · Worksheet with only exercise

4.4.1. This is an exercise's statement.

4.6 · Worksheet with both exercise and project

4.6.1. This is an exercise's statement.

4.12 Project answers, worksheet statement

4.2 · Subsection with only project

Project 4.2 Answer. This is a project's answer.

4.3 · Subsection with both exercise and project

Project 4.4 Answer. This is a project's answer.

4.4 · Worksheet with only exercise

4.4.1. This is an exercise's statement.

4.5 · Worksheet with only project

Project 4.5 Answer. This is a project's answer.

4.6 · Worksheet with both exercise and project

4.6.1. This is an exercise's statement.

Project 4.6 Answer. This is a project's answer.

4.13 Inline answers, project statement, worksheet solution

4.1 · Subsection with only exercise

Checkpoint 4.1 Answer. This is an exercise's answer.

4.2 · Subsection with only project

Project 4.2 This is a project's statement.

4.3 · Subsection with both exercise and project

Checkpoint 4.3 Answer. This is an exercise's answer.

Project 4.4 This is a project's statement.

4.4 · Worksheet with only exercise

4.4.1. Solution. This is the exercise's solution.

4.5 · Worksheet with only project

Project 4.5 This is a project's statement.

4.6 · Worksheet with both exercise and project

4.6.1. Solution. This is the exercise's solution.

Project 4.6 This is a project's statement.

5 Exercises, One Subsection

This <section> of the sample article demonstrates an “unstructured division.” There are no <subsection>, you are just reading the first two paragraphs, followed by some nonsense text. Then there is a *single* <exercises> division. Note that this division is not numbered (since it is unique within the <section>). And a cross-reference to one of the contained <exercise> will be numbered as a member of the <section>, [Exercise 5.4](#).

If you use the unstructured form of a division, *and* have both inline and divisional exercises, there is a potential to form ambiguous cross-references. To wit, check that [5.2](#) and [5.2](#) are really different exercises (which you are *unable to do* if you are reading this in print!). The solution is to include the type of exercise in the reference, which will assist everybody, but especially your print readers: [Checkpoint 5.2](#) and [Exercise 5.2](#).

The following text is mostly nonsense, just for testing purposes.

Checkpoint 5.1 Inline One. Aliquam vitae risus placerat, pellentesque leo vitae, iaculis ante. Praesent ac odio eget mi bibendum eleifend ac eget metus. Morbi in dolor et diam accumsan mattis. Aenean elementum pulvinar efficitur. Etiam viverra ut tellus quis consequat. Phasellus sit amet nisl a ligula pharetra tempus id in elit. Maecenas congue quam eu purus fermentum pretium. Fusce pellentesque ultricies arcu, egestas sollicitudin erat condimentum non. Integer non velit at dolor dictum aliquam et rhoncus mauris. Sed nec nibh id nunc convallis tincidunt ut at ligula. Etiam elementum nisl eu erat dapibus rhoncus.

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Checkpoint 5.2 Inline Two. Ut porttitor neque a pharetra euismod. Vivamus ut metus pretium, placerat massa tempor, condimentum metus. Phasellus vestibulum iaculis turpis non posuere. Vestibulum quis aliquet neque. Donec nec metus iaculis, laoreet massa vitae, suscipit tellus. Etiam et ultrices quam, quis pretium ligula. In ut cursus metus. Aenean volutpat quam odio, quis tempus dolor egestas eget. Nunc fringilla lobortis nunc, ut interdum lorem posuere sed. Sed sodales risus a laoreet venenatis. Nunc sodales tempor mollis. Nam sollicitudin velit sed ex viverra feugiat. Nunc consectetur mi vitae urna sollicitudin malesuada. Fusce eget risus lectus. Mauris augue velit, vestibulum vitae tempus sit amet, porttitor eget turpis.

Theorem 5.3 Major Result. *Vivamus tortor tortor, lobortis et sem vel, accumsan placerat libero. Sed eget metus non magna accumsan efficitur a non turpis. Curabitur maximus arcu ipsum, eget vestibulum nulla mollis ac. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Duis vel eleifend risus. Morbi hendrerit tellus eget nibh imperdiet, ac mollis nisl sagittis. Ut commodo pharetra leo. Suspendisse consequat velit eget velit condimentum feugiat.*

Checkpoint 5.4 Inline Three. Suspendisse lacinia mattis risus, eget viverra urna dictum eu. Maecenas ut sem in turpis egestas varius nec at ipsum. Praesent bibendum nisi et turpis congue, a pellentesque felis tempor. Vivamus non

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Exercise Collection

- Drill One.** Lorem ipsum dolor sit amet, consectetur adipiscing elit. Cras congue urna nulla. Aliquam eget euismod tellus. Maecenas nibh libero, venenatis a laoreet in, tempor sit amet sem. Morbi sit amet justo tempor velit auctor placerat. Maecenas nec lobortis orci. Aenean dictum enim lacus, ac blandit lacus elementum nec. Mauris porttitor neque volutpat tincidunt sollicitudin. Cras porta lectus ac facilisis tempor. Suspendisse in velit nisl. Sed convallis leo at nunc aliquet fermentum. Pellentesque feugiat at ex sed elementum. In porta vulputate ipsum sit amet consectetur.



Figure 5.5 A figure in an <exercise> in an *unstructured* division to test that it is numbered as if the containing <exercises> is not present



(a)
First
Panel,
subcaptioned



(b)
Second
Panel,
subcaptioned

Figure 5.6 A side-by-side in a figure in an <exercise> in an *unstructured* division to test that it is numbered as if the containing <exercises> is not present and to test the numbering of the panels in a solutions manual



Figure 5.7
First
Panel,
not
subcaptioned



Figure 5.8
Second
Panel,
not
subcaptioned

- Drill Two.** Aliquam sagittis ex at magna porttitor, quis scelerisque ligula malesuada. Vestibulum vitae mauris id nisi pretium ornare a id lectus. Nam suscipit magna id sem ultricies vestibulum. Nulla facilisi. Duis venenatis lectus massa, ac mollis nisi suscipit non. Sed et sapien vulputate, imperdiet nibh vel, lobortis nisl. Curabitur sagittis justo nibh. Nulla vulputate sodales justo at efficitur. Cras justo augue, mollis vitae nulla sed, vestibulum tincidunt ante. Vestibulum et lorem lectus. Maecenas accumsan lacus a nisi euismod rutrum eu dapibus justo. Etiam scelerisque a odio a euismod. Pellentesque vestibulum pellentesque nisi, a tempus erat cursus ac. Morbi ut auctor lectus.

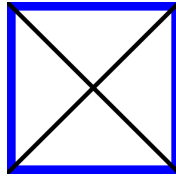


Figure 5.9 Testing numbering, figure in a <exercise> in an un-numbered <exercises>

3. **Drill Three.** Nam congue ex nec justo iaculis maximus. Vestibulum lobortis magna sed urna auctor, vel dignissim massa posuere. In sed venenatis elit. Vivamus congue gravida tempus. Nunc quis fermentum nisi. Nullam hendrerit lorem et tellus semper, sit amet scelerisque purus lobortis. Quisque sollicitudin quis neque eu suscipit. Praesent volutpat justo quis magna vehicula molestie.
4. **Challenging One.** Lorem ipsum dolor sit amet, consectetur adipiscing elit. Cras congue urna nulla. Aliquam eget euismod tellus. Maecenas nibh libero, venenatis a laoreet in, tempor sit amet sem. Morbi sit amet justo tempor velit auctor placerat. Maecenas nec lobortis orci. Aenean dictum enim lacus, ac blandit lacus elementum nec. Mauris porttitor neque volutpat tincidunt sollicitudin. Cras porta lectus ac facilisis tempor. Suspendisse in velit nisl. Sed convallis leo at nunc aliquet fermentum. Pellentesque feugiat at ex sed elementum. In porta vulputate ipsum sit amet consectetur.

Exercise Group. An introduction to an exercisegroup. This is here to attempt to interrupt the flow of the counting from this division to the next.

5. **Challenging Two.** Aliquam sagittis ex at magna porttitor, quis scelerisque ligula malesuada. Vestibulum vitae mauris id nisi pretium ornare a id lectus. Nam suscipit magna id sem ultricies vestibulum. Nulla facilisi. Duis venenatis lectus massa, ac mollis nisi suscipit non. Sed et sapien vulputate, imperdiet nibh vel, lobortis nisl. Curabitur sagittis justo nibh.
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6. **Challenging Three.** Nam congue ex nec justo iaculis maximus. Vestibulum lobortis magna sed urna auctor, vel dignissim massa posuere. In sed venenatis elit. Vivamus congue gravida tempus. Nunc quis fermentum nisi. Nullam hendrerit lorem et tellus semper, sit amet scelerisque purus lobortis. Quisque sollicitudin quis neque eu suscipit. Praesent volutpat justo quis magna vehicula molestie.
7. **Impossible One.** Lorem ipsum dolor sit amet, consectetur adipiscing elit. Cras congue urna nulla. Aliquam eget euismod tellus. Maecenas nibh libero, venenatis a laoreet in, tempor sit amet sem. Morbi sit amet justo tempor velit auctor placerat. Maecenas nec lobortis orci. Aenean dictum enim lacus, ac blandit lacus elementum nec. Mauris porttitor neque volutpat tincidunt sollicitudin. Cras porta lectus ac facilisis tempor. Suspendisse in velit nisl. Sed convallis leo at nunc aliquet fermentum. Pellentesque feugiat at ex sed elementum. In porta vulputate ipsum sit amet consectetur.

8. **Impossible Two.** Aliquam sagittis ex at magna porttitor, quis scelerisque ligula malesuada. Vestibulum vitae mauris id nisi pretium ornare a id lectus. Nam suscipit magna id sem ultricies vestibulum. Nulla facilisi. Duis venenatis lectus massa, ac mollis nisi suscipit non. Sed et sapien vulputate, imperdiet nibh vel, lobortis nisl. Curabitur sagittis justo nibh. Nulla vulputate sodales justo at efficitur. Cras justo augue, mollis vitae nulla sed, vestibulum tincidunt ante. Vestibulum et lorem lectus. Maecenas accumsan lacus a nisi euismod rutrum eu dapibus justo. Etiam scelerisque a odio a euismod. Pellentesque vestibulum pellentesque nisi, a tempus erat cursus ac. Morbi ut auctor lectus.
9. **Impossible Three.** Nam congue ex nec justo iaculis maximus. Vestibulum lobortis magna sed urna auctor, vel dignissim massa posuere. In sed venenatis elit. Vivamus congue gravida tempus. Nunc quis fermentum nisi. Nullam hendrerit lorem et tellus semper, sit amet scelerisque purus lobortis. Quisque sollicitudin quis neque eu suscipit. Praesent volutpat justo quis magna vehicula molestie.

More Reading

Left intentionally blank, just checking sectioning.

- [1] David C. Lay, *Subspaces and Echelon Forms*. The College Mathematics Journal, January 1993, **24** no. 1, 57–62.

6 Exercise Section, Structured

A collection of exercises, in a top-level <exercises> division, structured with <subexercises>.

Easy Exercises

1. **Arithmetic.** Compute $3 + 4$.
Answer. 7
2. **Trigonometry.** Compute $\sin(\pi/4)$.

Hard Problems

3. **Number Theory.** Prove Fermat’s Last Theorem.
4. **Millenial.** Find general solutions to the Navier-Stokes equation.

With an Exercise Group

Exercise Group. This is an exercise group, and this is its introduction.
And a second line for the introduction.

5. **One.** $6 - 5$

Hint. Subtract.

6. **Two.** $1 + 1$

Here we conclude the group.

And a second line for the conclusion.

7. **Outside exercisegroup, inside subexercises.** $6+5$